

Methodology Summary

Skill-Clustered Network Visualiser — Phase 1 (Days 0–8)

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Correction note — 6 October 2026 (D-77). This document records Phase 1, and it does not describe the method the project uses. Every figure in it was earned on 83 generated profiles built to be clusterable. That includes all three that §11 cites as validation evidence: silhouette 0.4745, HDBSCAN modularity 0.5224 and ARI 0.8985.

On the real network the convergence did not hold. Over 430 people, the agreement between greedy communities and HDBSCAN (ARI) fell to 0.047, and HDBSCAN’s modularity to 0.3352; over 442 people they were 0.0557 and 0.3429. HDBSCAN was discarded at the Phase 1.5 close-out on 19 July 2026, in favour of classification against the SOC taxonomy. So:

- §10’s conclusion that “the structure is real” holds for the generated data only.
- §11’s ledger, which lists HDBSCAN as chosen, records a decision that was later reversed.
- The “complete and validated” engine of §13 was only partly carried forward. The embeddings, projection, graph and centrality were; the clustering and its labels were not.

See `docs/phase1_5_methodology.md`, which reports these Phase 1 figures and makes their collapse on real data its central lesson. The run that produced the figures here survives only in the private repository’s history. The document below is left as written.

About this document

This is the readable, decision-led account of Phase 1 of the Skill-Clustered Network Visualiser — the “Core ML Engine” that turns a set of professional profiles into a validated, explorable network. It is written for a general technical reader and deliberately keeps the mathematics light: the aim is to explain **what was built, why each design choice was made, what the alternatives were, and how each stage was shown to work**. A companion document, *Mathematical Methodology*, gives the formal definitions and equations for every technique named here.

The single idea to carry through the whole thing: at every step we make a choice, name the credible alternative we rejected, and back the decision with a number the pipeline actually produced.

1. What the engine does, in one paragraph

The engine reads each person’s experience and turns it into a numerical “fingerprint” of meaning; groups people whose fingerprints are alike; lays the whole set out as a two-dimensional map; names each group automatically; connects similar people into a network; and finally measures who occupies the important positions in that network. Every one of these steps is validated on data whose correct answer is known in advance, so the results can be graded rather than merely admired.

2. Five principles behind every decision

Before the individual stages, it helps to state the design philosophy, because the same five commitments recur throughout and explain most of the choices.

Prove it on a known answer first. The whole pipeline is developed on synthetic data whose “true” groups we planted ourselves. Only a pipeline that can recover a known structure earns trust on unknown data.

One consistent notion of similarity. A single measure of closeness — the *angle* between two fingerprints, called cosine similarity — is used from the first stage to the last. No stage silently redefines what “similar” means, which is a common and hard-to-diagnose source of error.

Every choice defended against a named alternative. For each algorithm there was an obvious, popular competitor. The document records which one, and the concrete reason it was passed over — not as an afterthought, but as the point.

Keep the signal others discard. Many tools force every data point into a tidy group. This engine deliberately preserves loners (people who fit no group) and isolates (people connected to no one above the similarity bar), because in a professional network an individualist or a bridge is information, not an error.

Reproducibility as a feature. Fixed random seeds, cached intermediate results, and a single canonical output mean a stranger can clone the repository and reproduce every number in this document.

3. A testbed with a known answer — synthetic data (Days 0–2)

The project’s founding decision was to **not** scrape LinkedIn — a route that breaches the platform’s terms, creates real data-protection obligations, and reads as a liability rather than an asset on an application. Instead the engine is fed from permitted sources, and developed against a synthetic dataset.

Choosing synthetic data first pays off twice. It sidesteps the compliance problem entirely, and — more importantly for the methodology — it gives us **ground truth**. We generated 83 fictional profiles drawn from seven distinct professional “archetypes”, then deliberately corrupted about 15% of the fields to make the data realistically messy. Because we know which archetype each profile came from, every later stage can be scored against the right answer. When the clustering later recovers seven groups that match the seven archetypes, that is evidence the engine works — not a coincidence to be hoped for.

4. Turning words into meaning — embeddings (Day 3)

A computer cannot compare “experience” directly, so each profile’s text is converted into a list of 384 numbers — coordinates in a “meaning space” where people with similar backgrounds land near one another, even when they describe themselves in different words. This is done with a pretrained language model (all-MiniLM-L6-v2).

The alternative was simple keyword counting (TF-IDF over the raw profiles) or a paid embedding service. Keyword counting was rejected because it treats “ML Engineer” and “Data Scientist” as unrelated — it matches spellings, not meaning. A paid API was rejected because a small, local, free model keeps the whole project runnable by anyone, anywhere, with no keys or costs — exactly what a shareable portfolio piece needs. The model is small and fast enough to run on a laptop, yet captures genuine semantic similarity.

This is also where the project’s consistent measure of closeness is introduced. Two people are judged alike by the **angle** between their fingerprints, not their magnitude — so a person who wrote a long profile and one who wrote a short profile are still recognised as similar if they point the same way in meaning space. Angle is the natural, stable choice in a space of this many dimensions, and it is the measure the model was trained to make meaningful.

5. Finding the natural groups — clustering (Day 4)

The central modelling decision of Phase 1 is how to group people, and here the engine uses **HDBSCAN** rather than the far more common **K-means**.

K-means has three properties that are wrong for this problem. It requires you to state the number of groups in advance — but discovering how many communities exist is part of the question. It assumes groups are round, evenly sized blobs — but real skill clusters are irregular. And it forces every single person into a group — so a genuine one-of-a-kind profile gets absorbed into whichever cluster is least wrong, polluting it.

HDBSCAN avoids all three. It finds dense regions of similar people and discovers the number of groups by itself; it copes with irregular shapes; and it is willing to label a person as an **outlier** rather than mis-file them. On our data it found **seven groups and flagged two outliers** (about 2.4% of people) — the two most idiosyncratic profiles left honestly ungrouped rather than forced.

To check the grouping is real rather than arbitrary, we use the **silhouette score**, which asks, for each person, whether they sit closer to their own group than to the nearest rival group. The engine scores **0.4745** — a clear, positive separation. For messy real-world text (as opposed to clean textbook data), a value in this range indicates coherent, well-separated groups.

6. Drawing the map — projection (Day 5)

A 384-dimensional cloud of points cannot be drawn, but a picture is essential both for the eventual dashboard and for sanity-checking the clusters by eye. So the engine flattens the fingerprints down to two dimensions using **UMAP**, chosen over the popular **t-SNE**.

The difference that matters is what each method tries to preserve. t-SNE is excellent at keeping *near* neighbours together but tends to scramble the *global* arrangement — so the relative positions of whole clusters become meaningless, and distance between clusters on the plot can’t be trusted. UMAP preserves both local neighbourhoods and the broader layout, so the map is faithful at both scales, and it is faster to compute. (Plain dimensionality reduction such as PCA was ruled out earlier still — it is too crude to capture the non-linear structure of meaning.)

A map can lie by placing strangers next to each other, so we measure exactly that with a score called **trustworthiness**, which counts how often a projected neighbour was *not* really a neighbour in the original space. The engine scores **0.9840** out of 1 — the map almost never introduces false neighbours, which is what licenses us to reason about the clusters from the picture itself.

7. Naming the groups — labelling (Day 6)

Groups are useless if a human has to inspect them to know what they are, so the engine labels each cluster automatically by finding the words that are common *inside* that cluster but rare across the others — the terms that make a group distinctive. This recovered clean, human-readable names for all seven groups, for example *Engineering / Ansys / Engineer*, *Drug / Pharmacology / Clinical*, *Design / Interior*, and *VBA / Portfolio / Finance*.

This step quietly closes the loop opened in Day 0: the seven groups the engine discovered, entirely from the text, are the same seven archetypes we originally planted. The pipeline found what was really there.

8. From similarity to a network — the graph (Day 7)

Grouping tells us *who is alike*; a network tells us *how everyone relates* — who bridges between groups, who sits at the centre, who stands apart. So the engine connects the people into a graph, drawing a link between two people whenever their similarity clears a chosen bar.

Rather than an arbitrary rule, that bar is set by a single, interpretable dial: we choose a **target average number of connections per person** (here, eight) and let the bar follow. This keeps the network dense enough to reveal structure and sparse enough to read, and the knob means something concrete. The result is 332 connections across the 83 people. Consistent with the outlier philosophy, four people whose every similarity falls below the bar are left as **isolates** rather than artificially connected — low affinity is treated as signal.

The validation here is the strongest evidence in the whole of Phase 1. We measure **modularity**, which asks whether there really are more connections *inside* the groups than random chance would produce; the engine scores **0.5224**, well into “real community structure” territory. But the decisive point is *agreement across independent methods*: a completely separate community-detection algorithm, working only from the network and knowing nothing about the original clustering, draws almost exactly the same group boundaries — an agreement score (Adjusted Rand Index) of **0.8985**, where 0 is chance and 1 is identical. In other words, the same groups emerge whether you cluster in meaning-space, or detect communities in the network — three routes to the same structure.

9. Measuring who matters — centrality (Day 8)

The final stage asks a different question of each person: not *which group are you in*, but *how important is your position*? Importance, though, has several distinct meanings, so the engine reports four complementary measures rather than pretending there is one:

- **Most-connected** (degree) — simply how many links a person has.
- **Broker** (betweenness) — how often a person sits on the shortest path between others, i.e. how much they bridge otherwise separate groups.
- **Prestige** (PageRank) — importance that flows from being connected to *other* important people, in the spirit of the algorithm behind web search.
- **Reach** (closeness) — how close a person is, on average, to everyone they can reach.

Two choices here are worth calling out, because both are forced by an honest property of the data: the network is not fully connected — it falls into 13 separate pieces, with four people entirely alone. That disconnection breaks the naïve versions of two measures. For prestige we use PageRank instead of the classical **eigenvector centrality**, because eigenvector centrality quietly collapses on a fragmented network — run on our graph it drove **51 of the 83 people (61%) to essentially zero**, a misleading answer for two-thirds of the network. PageRank, by contrast, remains meaningful and gives even the isolated people a small, sensible score. For reach

we use a corrected form of closeness (Wasserman–Faust) that stops a person trapped on a tiny two-person island from appearing as central as a genuine hub.

The most useful insight from this stage is that **broker and prestige are not the same thing**, which is exactly why both are reported. Some people score high on both — central within their group *and* bridging to others — and these dual-role individuals are the highest-value connectors in the network. Others are strong brokers without being prestige hubs, or vice versa. Reassuringly, prestige is spread across five different groups rather than monopolised by one, indicating a healthy network with a genuine anchor in each community. (Looking ahead commercially, those dual-role connectors are precisely what a future team-allocation product would want to surface automatically.)

10. Why the result can be trusted — convergent validation

The reason to believe Phase 1 is not any single score but that **four independent checks, using different mathematics and different assumptions, all agree**. This is the methodological heart of the project.

Lens	What it inspects	Score	The claim it supports
Silhouette	tightness/separation of groups	0.4745	the groups are internally coherent
Trustworthiness	honesty of the 2-D map	0.9840	the picture faithfully reflects the data
Modularity	community structure in the network	0.5224	groups coincide with dense network regions
Adjusted Rand Index	agreement of two independent methods	0.8985	different methods find the same groups

No one of these could be dismissed as an artefact of a particular technique, because they come at the data from genuinely different angles — the geometry of the points, the honesty of the projection, the structure of the network, and the agreement between two unrelated grouping algorithms. When four such different tests point the same way, “the structure is real” stops being an assertion and becomes a defensible conclusion. The UMAP map provides a fifth, visual confirmation that a human can read at a glance.

11. The decision ledger

The entire Phase-1 argument condensed into one table: for each decision, the option taken, the credible alternative rejected, and the evidence that vindicates it.

Decision	Chosen	Alternative rejected	Why, and the evidence
Data source	Synthetic + own export	Scraping LinkedIn	Compliant <i>and</i> gives known ground truth; all 7 archetypes recovered
Meaning representation	Local sentence-embedding model	Keyword counting / paid API	Captures meaning not spelling; free and portable

Decision	Chosen	Alternative rejected	Why, and the evidence
Similarity measure	Cosine (angle)	Euclidean (distance)	Length-invariant, stable in high dimensions; used consistently end-to-end
Clustering	HDBSCAN	K-means	Finds the number of groups, handles irregular shapes, flags outliers; silhouette 0.4745
Projection	UMAP	t-SNE / PCA	Preserves global <i>and</i> local structure, faster; trustworthiness 0.9840
Labelling	Distinctive-term extraction	Manual naming	Automatic and interpretable; recovered all 7 archetype names
Network edges	Thresholded by target degree	Fixed k-nearest-neighbour	One interpretable density dial; modularity 0.5224
Community check	Modularity + second method + agreement score	Single method	Cross-method corroboration; ARI 0.8985
Prestige metric	PageRank	Eigenvector centrality	Robust on a fragmented network; eigenvector zeroed 61% of nodes
Reach metric	Wasserman–Faust closeness	Naïve closeness	Correctly handles the 13 disconnected components

12. Engineering discipline and reproducibility

The engine is built as a chain of independent stages, each with a clean input and output, so any one can be tested or swapped without disturbing the others. Random seeds are fixed throughout, so results are identical on every run. The expensive step — computing the fingerprints — is cached, so the pipeline is cheap to re-run. And the whole of Phase 1 funnels into a **single canonical table** of 83 people with their group, map coordinates, and four importance scores, which is the sole input the Phase-2 dashboard will consume. A stranger can clone the repository and reproduce every figure in this document from the README.

13. What Phase 1 sets up

The Core ML Engine is now complete and validated. Everything downstream reuses it: the canonical people-table drives the interactive dashboard (Phase 2); the same fingerprints power the natural-language matcher that will answer questions like “who in this network suits a fintech startup?” (Phase 3); and the entire engine generalises, essentially unchanged, from a personal network to a business tool for team allocation and project staffing (the eventual Stage-2 product). The methodological care taken in Phase 1 — every choice justified, every stage validated from

multiple angles — is the foundation the rest of the project stands on.